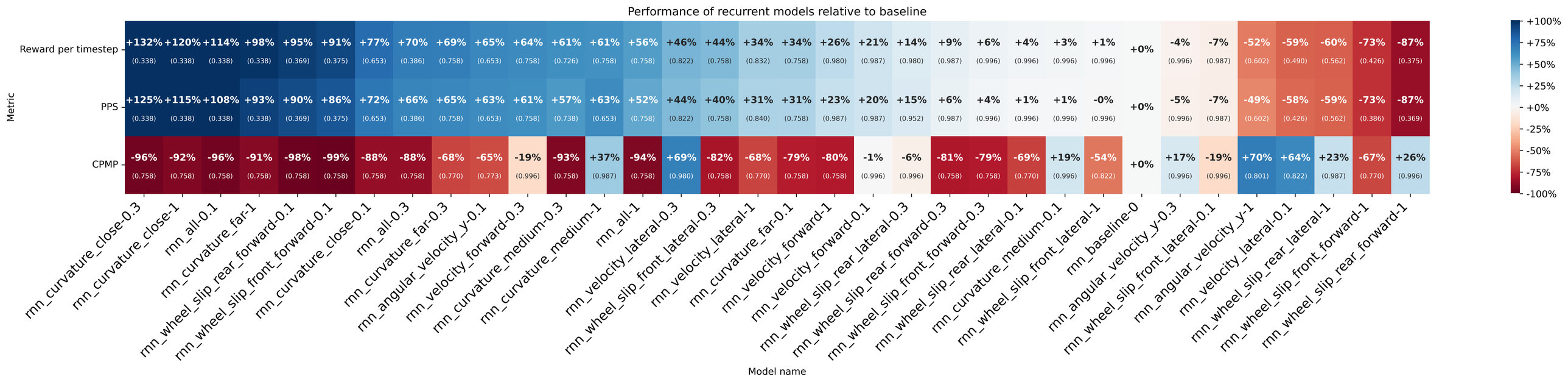
Performance of recurrent models relative to baseline
Metric
Reward per timestep
PPS
CPMP
+132% (0.338) +120% (0.338) +114% (0.338) +98% (0.338) +95% (0.369) +91% (0.375) +77% (0.653) +70% (0.386) +69% (0.758) +65% (0.653) +64% (0.758) +61% (0.726) +61% (0.758) +56% (0.758) +46% (0.822) +44% (0.758) +34% (0.832) +34% (0.758) +26% (0.980) +21% (0.987) +14% (0.980) +9% (0.987) +6% (0.996) +4% (0.996) +3% (0.996) +1% (0.996) +0% -4% (0.996) -7% (0.987) -52% (0.602) -59% (0.490) -60% (0.562) -73% (0.426) -87% (0.375)
+125% (0.338) +115% (0.338) +108% (0.338) +93% (0.338) +90% (0.369) +86% (0.375) +72% (0.653) +66% (0.386) +65% (0.758) +63% (0.653) +61% (0.758) +57% (0.738) +63% (0.653) +52% (0.758) +44% (0.822) +40% (0.758) +31% (0.840) +31% (0.758) +23% (0.987) +20% (0.987) +15% (0.952) +6% (0.987) +4% (0.996) +1% (0.996) +1% (0.996) -0% (0.996) +0% -5% (0.996) -7% (0.987) -49% (0.602) -58% (0.426) -59% (0.562) -73% (0.386) -87% (0.369)
-96% (0.758) -92% (0.758) -96% (0.758) -91% (0.758) -98% (0.758) -99% (0.758) -88% (0.758) -88% (0.758) -68% (0.770) -65% (0.773) -19% (0.996) -93% (0.758) +37% (0.987) -94% (0.758) +69% (0.980) -82% (0.758) -68% (0.770) -79% (0.758) -80% (0.758) -1% (0.996) -6% (0.996) -81% (0.758) -79% (0.758) -69% (0.770) +19% (0.996) -54% (0.822) +0% +17% (0.996) -19% (0.996) +70% (0.801) +64% (0.822) +23% (0.987) -67% (0.770) +26% (0.996)
rnn_curvature_close-0.3
rnn_curvature_close-1
rnn_all-0.1
rnn_curvature_far-1
rnn_wheel_slip_rear_forward-0.1
rnn_wheel_slip_front_forward-0.1
rnn_curvature_close-0.1
rnn_all-0.3
rnn_curvature_far-0.3
rnn_angular_velocity_y-0.1
rnn_velocity_forward-0.3
rnn_curvature_medium-0.3
rnn_curvature_medium-1
rnn_all-1
rnn_velocity_lateral-0.3
rnn_wheel_slip_front_lateral-0.3
rnn_velocity_lateral-1
rnn_curvature_far-0.1
rnn_velocity_forward-1
rnn_velocity_forward-0.1
rnn_wheel_slip_rear_lateral-0.3
rnn_wheel_slip_rear_forward-0.3
rnn_wheel_slip_front_forward-0.3
rnn_wheel_slip_rear_lateral-0.1
rnn_curvature_medium-0.1
rnn_wheel_slip_front_lateral-1
rnn_baseline-0
rnn_angular_velocity_y-0.3
rnn_wheel_slip_front_lateral-0.1
rnn_angular_velocity_y-1
rnn_velocity_lateral-0.1
rnn_wheel_slip_rear_lateral-1
rnn_wheel_slip_front_forward-1
rnn_wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%
Performance of feedforward models relative to baseline
Metric
Reward per timestep
PPS
CPMP
+14% (0.278) +7% (0.395) +6% (0.480) +3% (0.695) +2% (0.822) +1% (0.931) +0% -1% (0.931) -5% (0.639) -6% (0.480) -7% (0.546) -7% (0.423) -9% (0.540) -10% (0.429) -11% (0.429) -12% (0.395) -15% (0.480) -15% (0.329) -15% (0.378) -15% (0.356) -16% (0.334) -19% (0.334) -20% (0.334) -20% (0.395) -21% (0.159) -22% (0.392) -23% (0.255) -25% (0.334) -25% (0.159) -28% (0.159) -29% (0.356) -31% (0.334) -31% (0.334) -36% (0.159) -38% (0.107) -40% (0.278) -59% (0.159) -63% (0.159)
+13% (0.278) +7% (0.395) +5% (0.480) +3% (0.710) +2% (0.824) +1% (0.931) +0% -1% (0.931) -5% (0.639) -6% (0.480) -7% (0.545) -7% (0.423) -9% (0.540) -10% (0.429) -10% (0.436) -11% (0.395) -14% (0.480) -14% (0.329) -14% (0.374) -15% (0.356) -14% (0.334) -18% (0.338) -19% (0.334) -20% (0.395) -21% (0.159) -22% (0.392) -23% (0.255) -24% (0.334) -24% (0.159) -27% (0.159) -28% (0.356) -30% (0.334) -30% (0.334) -36% (0.159) -38% (0.107) -39% (0.278) -58% (0.159) -62% (0.159)
-64% (0.334) -47% (0.395) -62% (0.340) -54% (0.368) -32% (0.502) -33% (0.532) +0% +30% (0.768) +25% (0.732) +22% (0.662) +4% (0.934) -42% (0.423) +121% (0.480) +118% (0.480) +285% (0.392) +98% (0.340) +250% (0.423) +142% (0.352) +617% (0.480) +63% (0.374) +463% (0.334) +383% (0.356) +467% (0.443) +680% (0.395) +154% (0.278) +503% (0.472) +149% (0.280) +599% (0.374) +391% (0.159) +217% (0.334) +1614% (0.395) +405% (0.340) +451% (0.395) +91% (0.561) -52% (0.378) +1439% (0.423) +2440% (0.340) +1313% (0.340)
curvature_close-1
curvature_close-0.3
wheel_slip_front_forward-0.1
all-0.1
pcgrad_all-0.1
wheel_slip_rear_forward-0.1
baseline-0
velocity_forward-1
wheel_slip_front_lateral-1
all-0.3
pcgrad_all-0.03
wheel_slip_rear_forward-0.3
angular_velocity_y-1
curvature_close-0.1
curvature_far-0.1
curvature_medium-1
curvature_medium-0.1
pcgrad_all-0.3
wheel_slip_front_forward-0.3
curvature_medium-0.3
curvature_far-1
velocity_lateral-0.3
velocity_forward-0.3
velocity_forward-0.1
wheel_slip_rear_lateral-0.1
velocity_lateral-1
wheel_slip_front_lateral-0.1
wheel_slip_front_lateral-0.3
curvature_far-0.3
angular_velocity_y-0.1
wheel_slip_rear_lateral-1
wheel_slip_rear_lateral-0.3
velocity_lateral-0.1
pcgrad_all-1
all-1
angular_velocity_y-0.3
wheel_slip_front_forward-1
wheel_slip_rear_forward-1
Model name
+100%
+75%
+50%
+25%
+0%
-25%
-50%
-75%
-100%